Kartikeya Goel

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Education

Stanford University Expected Jun 2029
B.S. Computer Science; Minor in Statistics — GPA: 3.66 *Stanford, CA*

- **Coursework:** CS 106B (Data Structures & Algorithms), CS 107 (Computer Organization & Systems), STATS 117 (Probability), MATH 51 (Linear Algebra & Multivariable Calculus), MATH 104 (Matrix Theory)
- **Fall 2026:** CS 103, CS 109, CS 161, CS 229

Experience

Apple Jun 2026 – Present
Software Engineering Intern, Marketing Communications *Sunnyvale, CA*

- Building internal GenAI developer tooling that integrates into production screen workflows rather than a chat interface; model calls are wrapped in deterministic pre- and post-processing so outputs are schema-validated before entering the pipeline.
- Shipped a localization tool replacing a manual per-screen process across 37 locales, owned end to end from design through deployment; now used by 50+ designers and producers.

Sago Jan 2026 – Present
Software Engineering Intern *Remote*

- Own brief generation end to end: a service ingesting company and market data, drafting structured investment briefs, and serving them as conversational reports queried live in deal meetings.
- Built the retrieval and citation layer so each generated claim resolves to a source document.

MARi Jun 2025 – Aug 2025
Software Engineering Intern *Remote*

- Built the internal eval harness for an automated decisioning pipeline; identified a prompt-injection failure mode and raised pass rate from 23% to 95% on a 37-case suite.
- Parallelized Python processing over multi-GB inputs, reducing a 17-hour job to 30 minutes.

IQ Spectra Inc. Mar 2023 – Apr 2025
Software Engineering Intern (part-time during school year) *Remote*

- Built web automation tooling extracting and normalizing data across 10,000+ sources, handling pagination, rate limits, and inconsistent markup.
- Cut end-to-end runtime 34% by re-keying the HTML cache on content hash so unchanged pages skipped re-parsing entirely, cutting peak memory roughly in half.
- Traced and fixed a memory leak in the parser using Valgrind and GDB.

Projects

FRC Competition Robot (Sole Programmer) Jan 2024 – Dec 2024
Java, Robotics, Autonomous Navigation, Real-Time Control *GitHub*

- Wrote the entire control stack for the team's competition robot in Java, including subsystem state machines, command scheduling, autonomous routines, and driver controls, on a 50 Hz control loop with hard timing budgets.
- Implemented pose estimation fusing wheel odometry, gyroscope, and vision measurements for autonomous localization and path following on the field.
- Debugged dropped control cycles caused by blocking calls in the main loop; restructured the superstructure as a non-blocking state machine to guarantee bounded per-cycle work.

Crucible (1st Place, Claude Buildathon) Apr 2026
Python, Multi-Agent Systems

- Built a deliberation tool running competing model agents against each other to surface disagreement before a recommendation; won 1st of 50+ teams.

Skills

Languages: Java, Python, C++, Bash, C#, TypeScript
Robotics & Systems: Real-time control, state machines, sensor fusion, pose estimation, concurrency, performance profiling
Data & Infrastructure: PostgreSQL, SQLite, Git, Docker, Linux, Make/CMake, GDB, Valgrind, AWS, Google Cloud
AI & ML: PyTorch, scikit-learn, LLM tooling, RAG, multi-agent pipelines, evaluation harnesses, AI-assisted development